

PROJECT DOCUMENTATION

UK Train Rides

Power BI Analytics & Dashboard Project

Graduation Project — Data Analytics Track

April 2026

1. Project Planning & Management

1.1 Project Overview

This project analyzes UK train ride data to generate actionable business insights using Microsoft Power BI. The dataset contains 31,653 ticket transactions spanning journeys from January to April 2024, covering ten major UK stations. The goal is to build an interactive dashboard that helps railway management understand revenue, punctuality, passenger behavior, and demand forecasting.

1.2 Project Scope

- Dataset: 31,653 train ticket transactions, 18 fields per record
- Date range covered: January 1, 2024 to April 30, 2024 (journeys); purchases as early as December 2023
- Stations covered: Manchester Piccadilly, London Euston, Liverpool Lime Street, London Paddington, London Kings Cross, London St Pancras, Birmingham New Street, York, Reading, Oxford
- Tool: Microsoft Power BI (data cleaning, modeling, visualization, and forecasting)
- Final deliverable: Interactive Power BI dashboard + project documentation + presentation

1.3 Project Timeline

Week	Phase	Key Tasks	Deliverable
Week 1	Data Cleaning & Preprocessing	Import data, fix data types, standardize values, handle nulls, create calculated columns	Cleaned dataset in Power BI
Week 2	Analysis Questions Phase	Define business questions, build data model, create DAX measures, forecast next month	Analysis questions + DAX measures
Week 3	Dashboard Phase	Design and build interactive Power BI dashboard with visuals, filters, slicers	Published Power BI Dashboard
Week 4	Final Presentation	Prepare documentation, presentation slides, and project summary	Report + Presentation

1.4 Team Roles & Responsibilities

Role	Responsibilities
Data Analyst	Data cleaning, DAX measure creation, exploratory data analysis
Dashboard Designer	Power BI report layout, visual selection, UX/interactivity
Project Manager	Timeline tracking, documentation, stakeholder communication
Presenter	Final report writing, slide preparation, presentation delivery

1.5 Tools & Technologies

- Microsoft Power BI Desktop — primary analysis and visualization tool
- Microsoft Excel / CSV — source data format
- DAX (Data Analysis Expressions) — for calculated measures and KPIs
- Power Query — for data transformation and cleaning
- GitHub — for version control and submission

2. Literature Review

2.1 Overview of Railway Analytics

Railway analytics has emerged as a critical area in transportation management. Organizations such as Network Rail (UK) and Transport for London (TfL) routinely use data analytics to optimize timetables, manage delays, and improve passenger experience. The rise of Business Intelligence (BI) tools has made it easier to transform raw transactional data into decision-ready dashboards.

2.2 Relevance of Power BI in Transport Analytics

Microsoft Power BI is widely adopted in the transport sector due to its ability to connect to large datasets, perform real-time refresh, and enable self-service analytics. It supports DAX calculations, time-intelligence functions, and built-in forecasting — all of which are directly applicable to this project.

2.3 Key Analytical Themes in Railway Data

2.3.1 Revenue & Ticket Analysis

Studies in transit revenue management show that ticket type mix (Advance, Off-Peak, Anytime) and class (Standard vs. First) are the primary drivers of revenue variability. Advance tickets, priced at 50% off, drive volume but reduce yield per seat. Understanding this trade-off is essential for revenue optimization.

2.3.2 On-Time Performance & Delay Analysis

Delay root-cause analysis is one of the most studied topics in railway operations research. Common causes include signal failures, technical issues, and weather. Studies show that reducing delay rates by even 5% significantly improves customer satisfaction scores and reduces refund-related revenue leakage.

2.3.3 Demand Forecasting

Time-series forecasting of passenger demand is a well-established practice. Power BI provides a built-in exponential smoothing forecasting model that can project future ride volumes based on historical patterns. This is relevant to the project requirement of predicting rides and revenue for the next month.

2.3.4 Customer Segmentation

Railcard segmentation (Adult, Senior, Disabled, None) represents a form of price discrimination commonly studied in transport economics. Understanding which segments use which ticket types and routes allows for targeted promotions and better resource allocation.

3. Requirements Gathering

3.1 Business Requirements

The following business requirements were identified based on the project brief and dataset:

#	Business Requirement	Priority
BR-01	Understand revenue breakdown by route, ticket type, ticket class, and time period	High
BR-02	Identify peak travel days, times, and routes	High
BR-03	Analyze on-time performance and delay causes across stations and routes	High
BR-04	Measure refund request rate and its relationship to delays/cancellations	Medium
BR-05	Forecast number of rides and total revenue for the next month (May 2024)	High
BR-06	Understand demand by ticket class (Standard vs. First Class)	Medium
BR-07	Analyze purchase behavior (online vs. station, payment methods)	Low
BR-08	Identify most and least profitable routes	Medium

3.2 Functional Requirements (Power BI Dashboard)

- The dashboard must include filters/slicers for date range, station, ticket class, and ticket type
- KPI cards must display: Total Revenue, Total Rides, On-Time Rate (%), Average Ticket Price, Refund Rate (%)
- A time-series line chart must show ride volume and revenue trends over time
- A bar/column chart must compare revenue and volume by departure station
- A matrix or heatmap must show journey status breakdown by route
- A forecast visual must project next month's rides and revenue using Power BI built-in forecasting
- A donut or pie chart must show ticket class demand split

3.3 Non-Functional Requirements

- The Power BI file (.pbix) must be saved and submitted via GitHub

- All data cleaning steps must be documented in Power Query or a preprocessing notebook
- DAX measures must be clearly named and organized in a dedicated Measures table
- The dashboard must be navigable without training — self-explanatory labels and tooltips

3.4 Data Requirements & Quality Issues

The following data quality issues were identified during preprocessing that must be addressed in Power BI Power Query:

Column	Issue Found	Resolution
Railcard	Null/missing values (~40% of records have no railcard)	Replace nulls with 'None'
Reason for Delay	Nulls for all On Time journeys; also inconsistent values ('Weather' vs 'Weather Conditions', 'Staffing' vs 'Staff Shortage', 'Signal Failure' vs 'Signal failure')	Replace nulls with 'N/A'; standardize text values
Date of Purchase / Date of Journey	Stored as text strings, not date types	Convert to Date type in Power Query
Departure Time / Arrival Time / Actual Arrival Time	Stored as text, not time	Convert to Time type
Price	Integer type — acceptable, but verify no negative values	Add validation filter
Journey Status	3 values: On Time, Delayed, Cancelled — clean	No action needed

4. System Analysis & Design

4.1 Data Model Design

The dataset is a single flat table (31,653 rows, 18 columns). In Power BI, we will restructure this into a star schema by splitting out dimension tables to improve performance and enable cleaner DAX measures.

4.1.1 Proposed Star Schema

Table	Type	Key Fields	Purpose
FactTransactions	Fact Table	Transaction ID, Date of Journey (FK), Price, Journey Status	Central fact table with all ticket purchases
DimDate	Dimension	Date (PK), Year, Month, Month Name, Day, Weekday, Is Weekend	Time intelligence — enables forecasting and time-series analysis
DimStation	Dimension	Station ID (PK), Station Name, Region	Route analysis and station-level filtering
DimTicket	Dimension	Ticket Key (PK), Ticket Class, Ticket Type, Purchase Type, Payment Method	Ticket segmentation analysis
DimRailcard	Dimension	Railcard (PK): None, Adult, Senior, Disabled	Customer segment analysis

4.2 Key DAX Measures

The following DAX measures will be created in Power BI:

Measure Name	DAX Formula (simplified)	Used In
Total Revenue	SUM(FactTransactions[Price])	KPI card, line chart, forecast
Total Rides	COUNTROWS(FactTransactions)	KPI card, line chart, forecast
On Time Rate %	DIVIDE(COUNTROWS(FILTER(Fact, [Journey Status]="On Time")), [Total Rides])	KPI card, gauge
Avg Ticket Price	AVERAGE(FactTransactions[Price])	KPI card, route comparison

Measure Name	DAX Formula (simplified)	Used In
Refund Rate %	<code>DIVIDE(COUNTROWS(FILTER(Fact, [Refund Request]="Yes")), [Total Rides])</code>	KPI card, delay analysis
Delay Rate %	<code>DIVIDE(COUNTROWS(FILTER(Fact, [Journey Status]="Delayed")), [Total Rides])</code>	Journey status chart
Revenue per Route	<code>CALCULATE([Total Revenue], route context)</code>	Bar chart by route
MoM Revenue Growth %	<code>([This Month Revenue] - [Last Month Revenue]) / [Last Month Revenue]</code>	Trend analysis

4.3 Analysis Questions Answered by the Dashboard

The following business questions are answerable using the dataset and will be visualized in the Power BI dashboard:

Revenue & Financial Analysis

- What is the total revenue generated per month?
- Which route generates the highest revenue?
- What is the revenue split between Standard and First Class tickets?
- How does revenue vary by ticket type (Advance vs. Off-Peak vs. Anytime)?

Demand & Volume Analysis

- Which departure stations have the most journeys?
- What are the peak travel days of the week?
- How does demand vary by hour of departure?
- Which ticket class (Standard / First Class) has higher demand?

Punctuality & Delay Analysis

- What percentage of journeys are On Time, Delayed, or Cancelled?
- What are the most common reasons for delays?
- Which routes have the worst on-time performance?
- What is the relationship between journey delays and refund requests?

Customer & Purchase Behavior

- What proportion of tickets are purchased online vs. at the station?
- Which payment method is most commonly used?

- What percentage of passengers hold a railcard?
- Do railcard holders prefer Advance or Off-Peak tickets?

Forecasting (Next Month — May 2024)

- Predicted total number of rides for May 2024 (using Power BI built-in time-series forecast)
- Predicted daily revenue for each day in May 2024
- Expected demand by ticket class in May 2024

4.4 Dashboard Layout Design

Page	Page Name	Visuals Included
1	Executive Summary	KPI cards: Total Revenue, Total Rides, On-Time Rate, Avg Price, Refund Rate; Revenue trend line chart; Rides by month bar chart
2	Route & Station Analysis	Map visual of stations; Revenue by route bar chart; Top 5 / Bottom 5 routes; Arrival vs. Departure station matrix
3	Ticket & Revenue Analysis	Revenue by ticket type (Advance/Off-Peak/Anytime); Revenue by ticket class; Purchase type split (Online vs. Station); Payment method donut chart
4	Punctuality & Delay Analysis	On-Time Rate gauge; Journey status donut chart; Delay reasons bar chart; Refund rate by delay reason
5	Forecasting	Rides forecast for May 2024 (line chart with Power BI forecast); Revenue forecast for May 2024; Ticket class demand forecast

5. Appendix: Dataset Summary

Field	Type	Description	Example Values
Transaction ID	Text (UUID)	Unique ticket purchase identifier	da8a6ba8-b3dc...
Date of Purchase	Date	Date ticket was bought	2023-12-08
Time of Purchase	Time	Time ticket was bought	12:41:11
Purchase Type	Category	Where ticket was bought	Online, Station
Payment Method	Category	Payment used	Contactless, Credit Card, Debit Card
Railcard	Category	Discount card held	Adult, Senior, Disabled, None
Ticket Class	Category	Seat class	Standard, First Class
Ticket Type	Category	Pricing rule	Advance, Off-Peak, Anytime
Price	Integer (GBP)	Final ticket price paid	1 – 267
Departure Station	Text	Origin station	Manchester Piccadilly
Arrival Destination	Text	Destination station	Birmingham New Street
Date of Journey	Date	Travel date	2024-01-01
Departure Time	Time	Scheduled departure	11:00:00
Arrival Time	Time	Scheduled arrival	13:30:00
Actual Arrival Time	Time	Actual arrival time	13:30:00
Journey Status	Category	Outcome of journey	On Time, Delayed, Cancelled
Reason for Delay	Text (nullable)	Cause of delay if applicable	Signal Failure, Weather...
Refund Request	Binary	Whether refund was claimed	Yes, No